Unit 3 Assignment 1

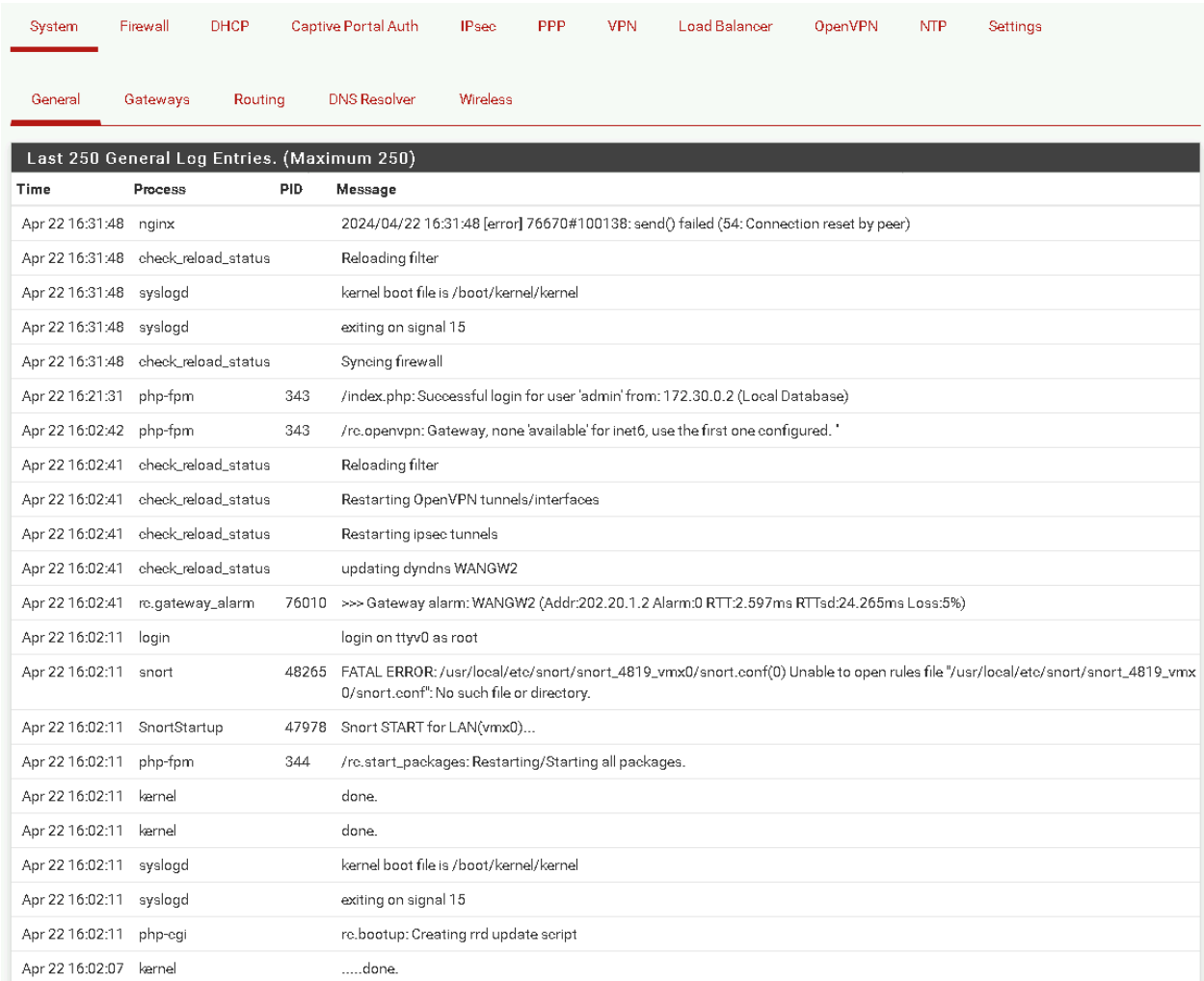
Name: Jonathan Hernandez

Lab Screenshots and Narrative

1. Insert and title all lab screenshot(s) in the order they were taken.

Section 1, Part 1

During this network security exercise, I configured the firewall to record a log of blocked connections. This simplifies log analysis and aids in identifying network traffic specific to the lab environment. In real-world deployments, logging allowed connections or all traffic might be more advantageous, depending on the organization's security posture and risk tolerance.



System | Firewall | DHCP | Captive Portal Auth | IPsec | PPP | VPN | Load Balancer | OpenVPN | NTP | Settings

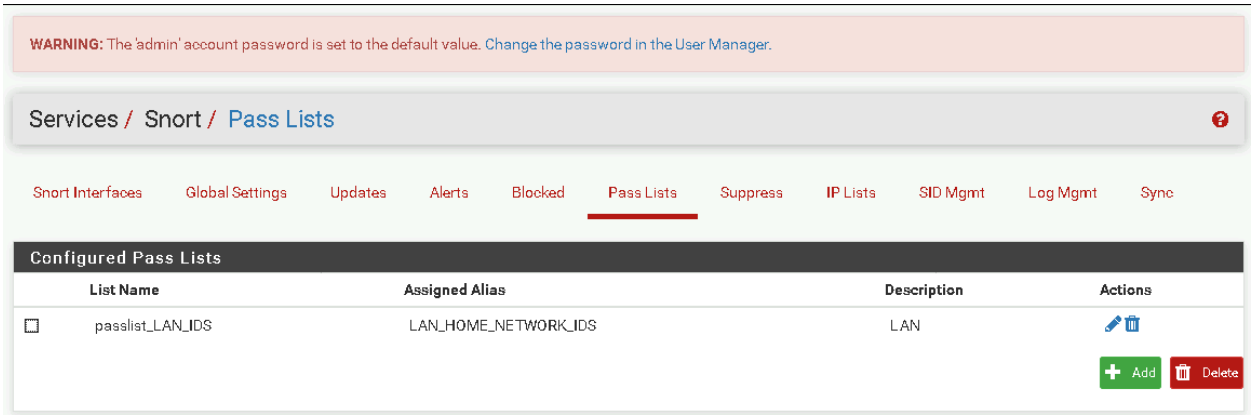
Normal View | Dynamic View | Summary View

Last 166 Firewall Log Entries. (Maximum 250)

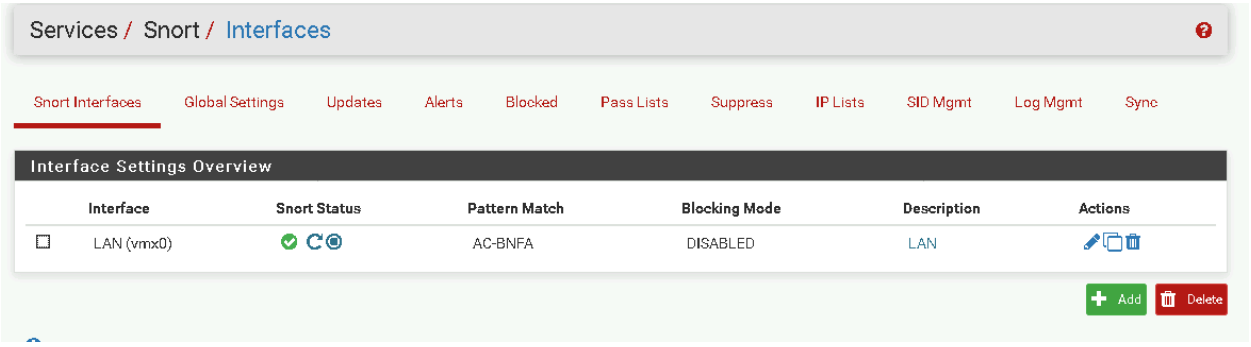
Action	Time	Interface	Source	Destination	Protocol
✖	Apr 22 16:33:10	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:10	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:09	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:08	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:08	WAN	202.20.1.1:34313	8.8.8.8:53	UDP
✖	Apr 22 16:33:08	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:07	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:07	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:06	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:06	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:05	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:05	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:04	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:04	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:03	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:03	WAN	202.20.1.1:34313	8.8.8.8:53	UDP
✖	Apr 22 16:33:03	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:02	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:02	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:01	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:00	WAN	202.20.1.2	202.20.1.1	ICMP
✖	Apr 22 16:33:00	WAN	202.20.1.2:50858	199.7.91.13:53	UDP

Section 1, Part 2

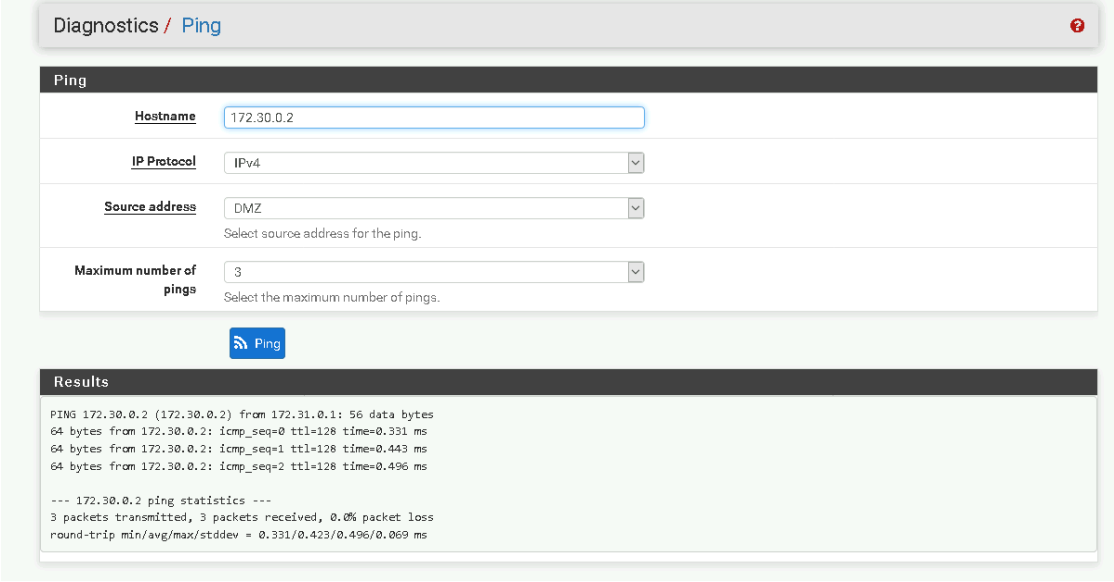
I configured Snort's intrusion detection capabilities by leveraging pre-defined rules that flag unusual network activity. Steps 1-5 involved creating an "internal source definition" (ISD) passlist based on a pre-configured pfSense alias for the local area network (LAN) within this lab environment. This passlist allows Snort to differentiate the source, destination, and direction of communication between monitored networks. This is crucial for accurate IDS rule evaluation, enabling Snort to distinguish "home" traffic from external traffic for effective anomaly detection.

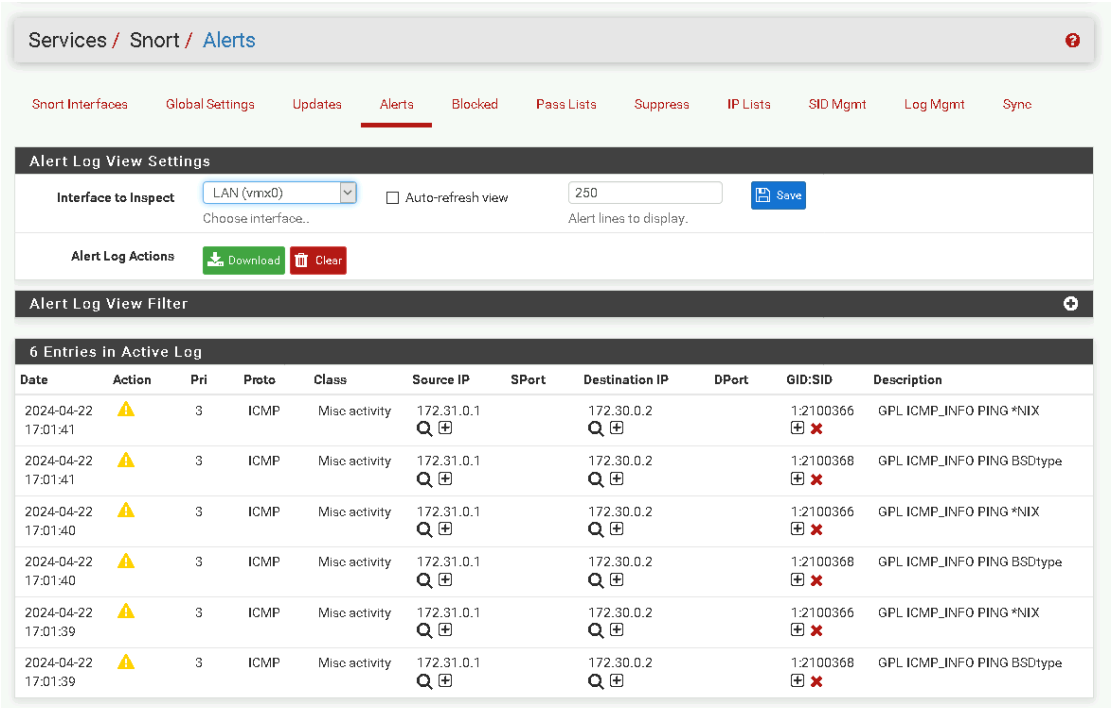


I reviewed the interface settings overview module. Upon clicking the play button, a green checkmark materialized, signifying successful Snort intrusion detection configuration on the LAN interface.



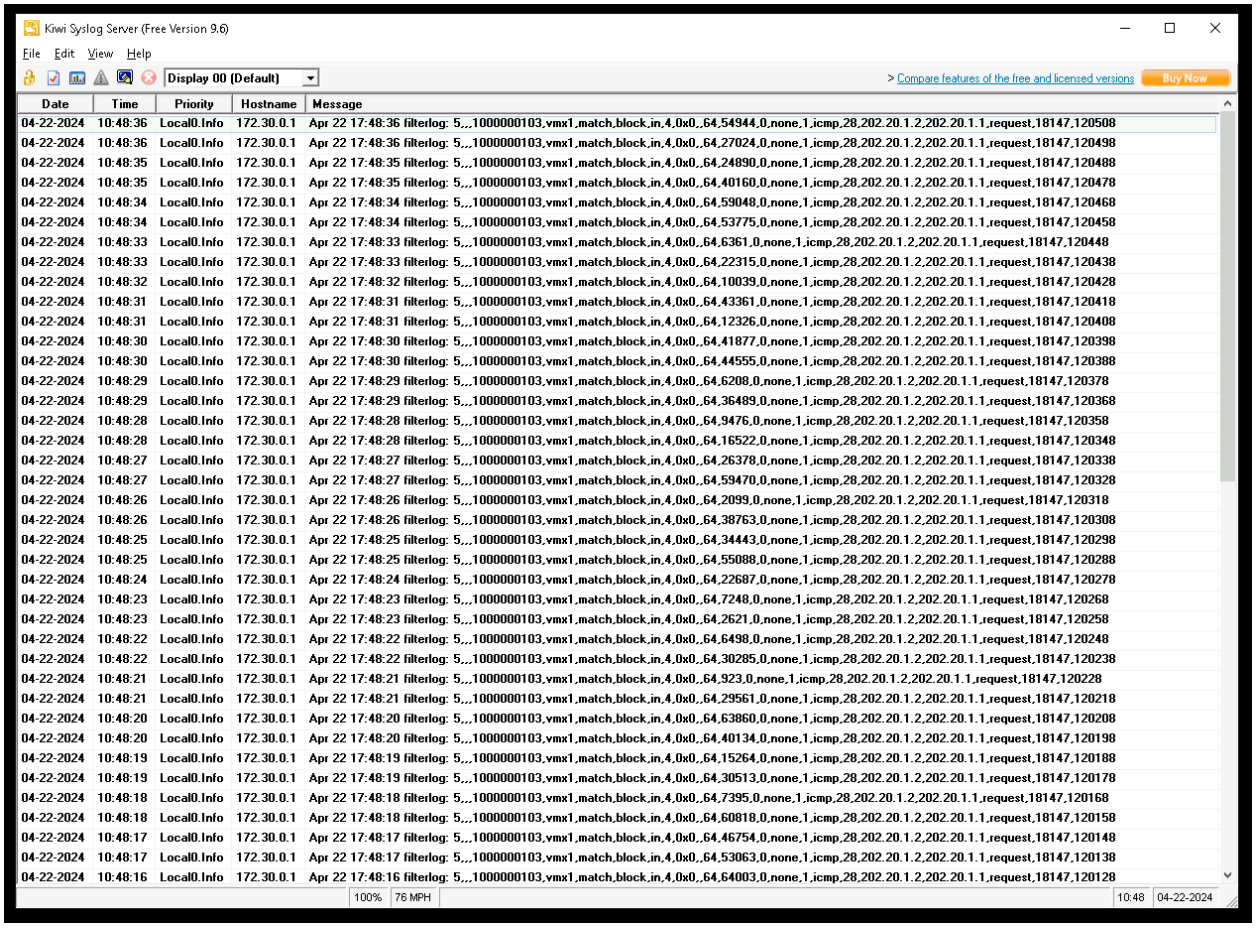
I proceeded to test Snort's monitoring capabilities against the defined ruleset. Recall that the ruleset configuration was completed earlier. If successful, Snort would initiate a ping towards a designated LAN IP, in this case, the virtual workstation at 172.30.0.2, and elicit an ICMP Echo Reply. Gratifyingly, the results module confirmed successful ping communication between the DMZ network and the target workstation, verifying Snort's ability to monitor for events matching the applied ruleset.



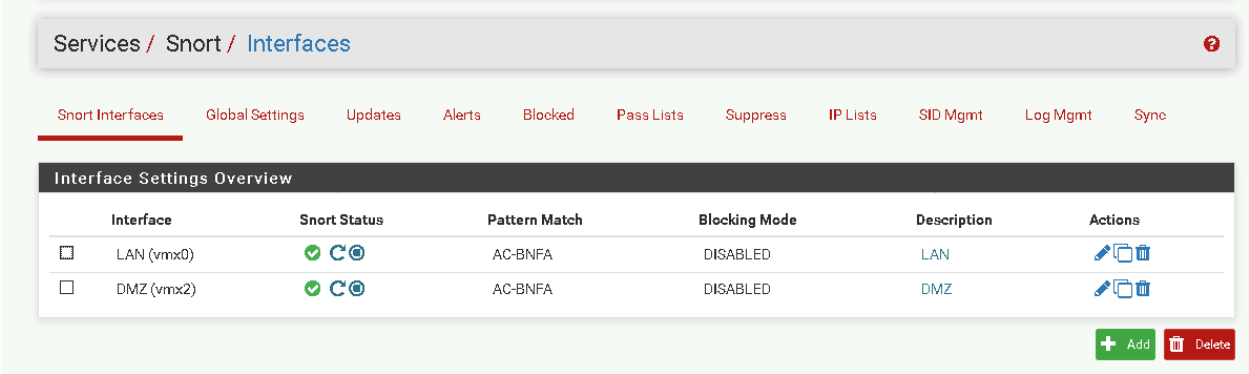


Section 1, Part 3

In this image, we have successfully setup Kiwi server. It has been configured so that the traffic from 172.30.0.1 is being monitored at all times. It will also send alerts if needed or assigned a ruleset.

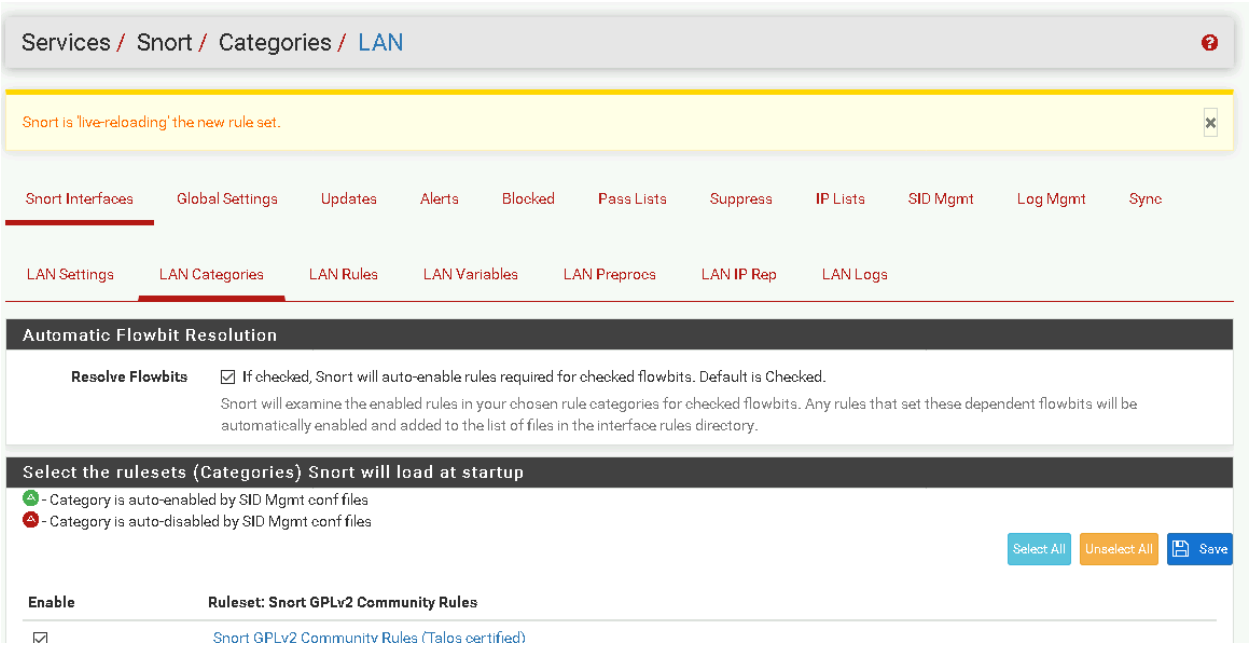


In this screenshot, I have successfully added the DMZ and LAN subnetwork interface so that it may be monitored from the Snort interface. After completion, it has successfully been activated and it is working correctly. Snort is now actively monitoring my organization's rDMZ subnetwork for events that align with the implemented rulesets. This enhanced visibility aids in the timely detection and mitigation of potential security threats.



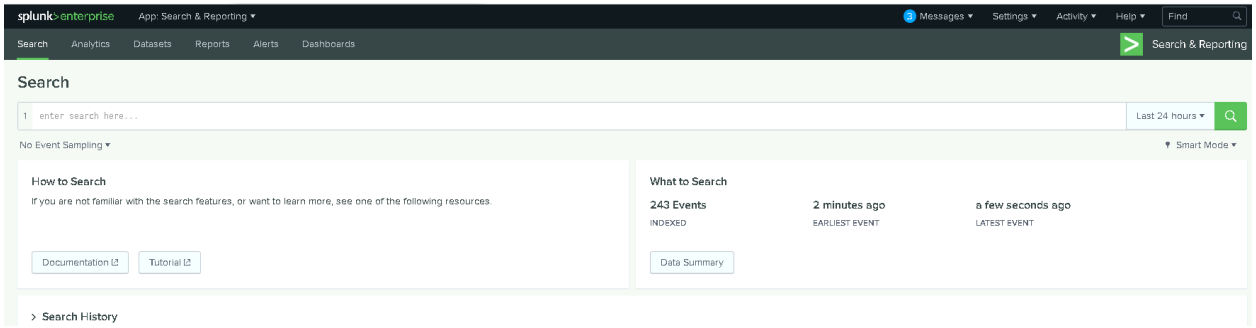
Section 2, Part 1

My Snort configuration reflects the distinct security requirements of the monitored subnetworks. The LAN, housing a diverse range of private systems, necessitated a more comprehensive ruleset compared to the DMZ, which solely accommodates an internet-of-Things web server. This tailored approach ensures optimal intrusion detection for each environment without unnecessary resource consumption



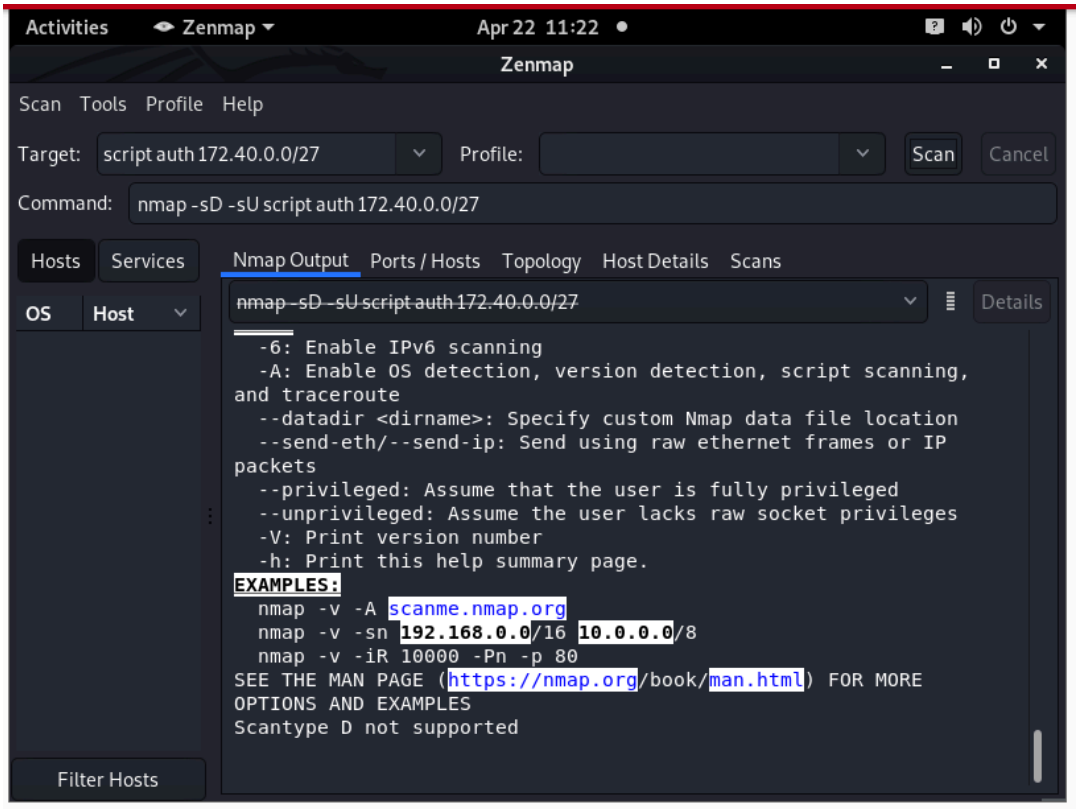
Section 2, Part 2

My Splunk configuration is operational, successfully receiving syslog datagrams forwarded from network systems. This seamless integration leverages the pre-configured log forwarding established on the pfSense firewall in section 1. As a result, Splunk is now be actively ingesting security logs, enabling me to leverage its powerful analysis capabilities.



Section 2, Part 3

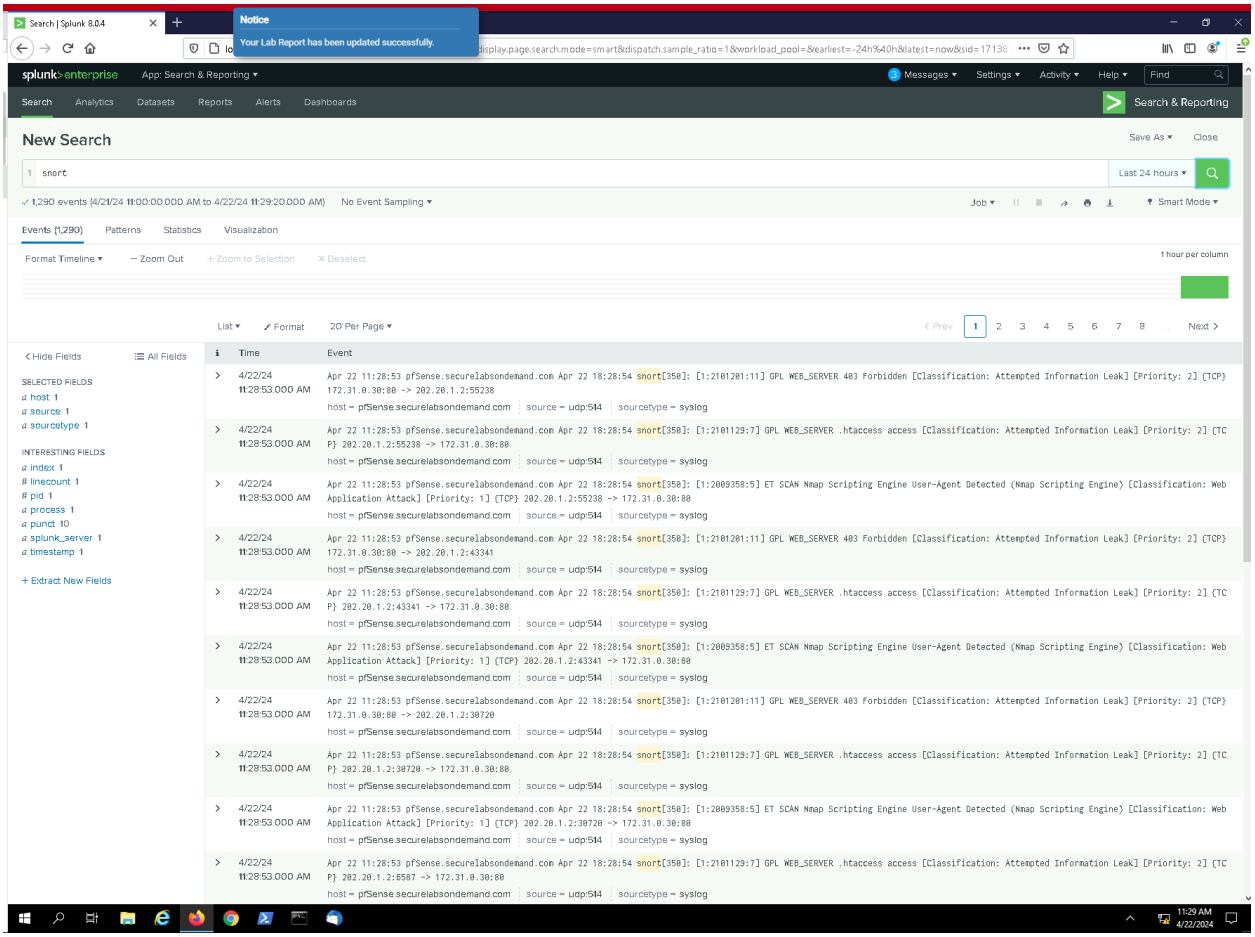
In this section, we have connected to our Attack Computer. From here, I was able to open Zenmap and use the command nmap -sS -sU - -script auth 172.40.0.0/27 to trace the WAN IP back to the configured DMZ and scan it for open ports.



After scanning the ports using the attack computer, we can see in Splunk that the traffic can be seen from the attack computer.

If there was a real-world scenario, we would be able to see in real-time and prevent any cyber attack from being made.

We can see from the screenshot that the necessary rules and configurations have been set to monitor the network.



Assignment

I have implemented a robust, layered security solution to safeguard my organization's network. By deploying Snort intrusion detection on a pfSense firewall, I established real-time monitoring for suspicious activity. Centralized log management through a Syslog server streamlines analysis, while Splunk SIEM integrates seamlessly for advanced threat analysis. This multi-faceted approach allows for tailored security – the LAN, with its diverse systems, employs a more comprehensive Snort ruleset compared to the DMZ's internet-of-things web server. Finally, successful penetration testing using Kali Linux and Infection Money validated the effectiveness of the implemented solution, giving me confidence in our ability to detect and respond to potential cyber threats. This enhanced security posture empowers my organization to proactively manage risks and ensure the continued protection of our valuable assets. Now, security analysts can leverage Splunk's powerful insights to investigate potential incidents with greater efficiency, further strengthening our overall cyber defense capabilities.